TECHNICAL IMPLEMENTATION REPORT

FACULTY AND STUDENT ACHIEVEMENT PORTAL

A Project Report on System Architecture, Authentication, Dashboard Functionality, Responsive Design, Navigation, Forms, and API Integration

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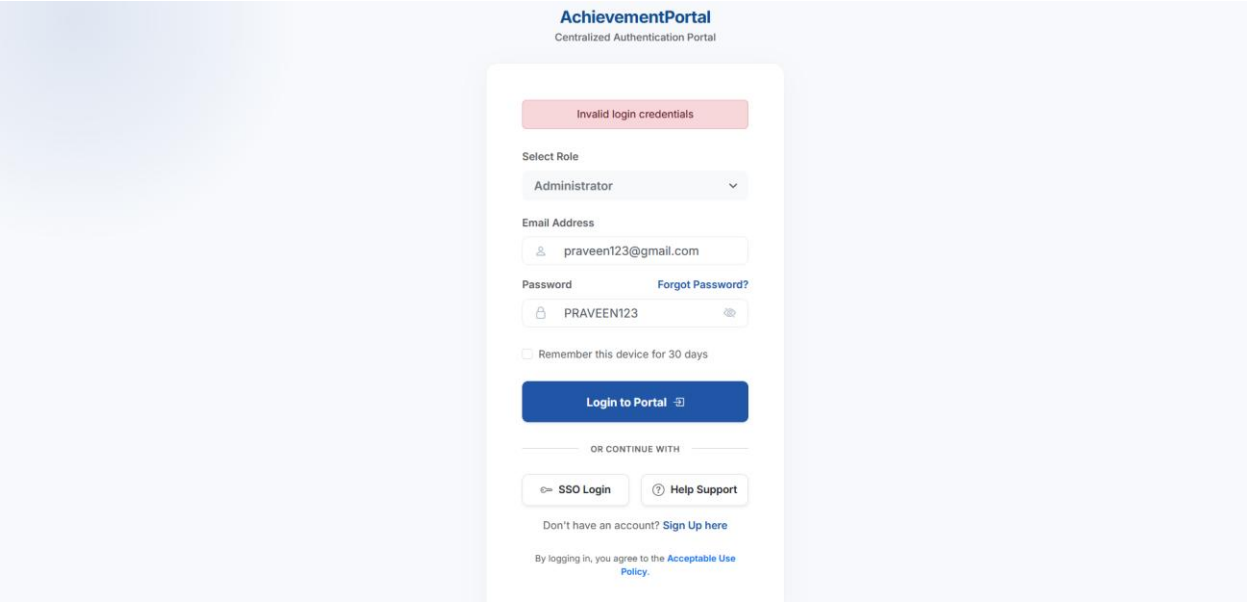
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1. Authentication

1.1 Introduction

Authentication constitutes the entry point through which every user of the College Portal establishes a verified identity before gaining access to role-specific functionality. In a multi-stakeholder academic environment involving students, faculty, and administrators, it is essential that the system be able to distinguish between these categories of users and grant privileges accordingly. The College Portal addresses this requirement through a centralized, multi-role authentication mechanism that combines identity verification with explicit role selection, ensuring that access to sensitive academic data and administrative operations is restricted to appropriately authorized individuals.

Rather than relying on a single generic login form, the system requires the user to declare their role at the point of authentication. This design choice reflects the underlying data model of the application, in which student, faculty, and administrator records are maintained in separate database tables. The authentication subsystem therefore performs the dual function of verifying credentials and confirming that the declared role is consistent with the user's actual record in the database.



1.2 Purpose

The primary purpose of the authentication module is to establish a secure and reliable gateway that prevents unauthorized access to the portal while simultaneously directing each

authenticated user to the dashboard and functionality appropriate to their role. Since the portal serves three distinct categories of users with materially different privileges, ranging from submitting achievements as a student to approving them as faculty or configuring institutional parameters as an administrator, it is imperative that role determination be enforced at the earliest possible stage of interaction with the system. The authentication module also serves the purpose of maintaining session continuity, so that once a user has been verified, subsequent navigation within the application does not require repeated credential submission.

1.3 Working Principle

The authentication workflow begins when a user navigates to the login page and selects their role, namely Student, Faculty, or Admin, from a dropdown control before entering their email address and password. Upon submission, the application first authenticates the supplied credentials against Supabase Auth's internal user store using the signInWithPassword method. This step confirms that the email and password combination corresponds to a valid, registered account, independent of role.

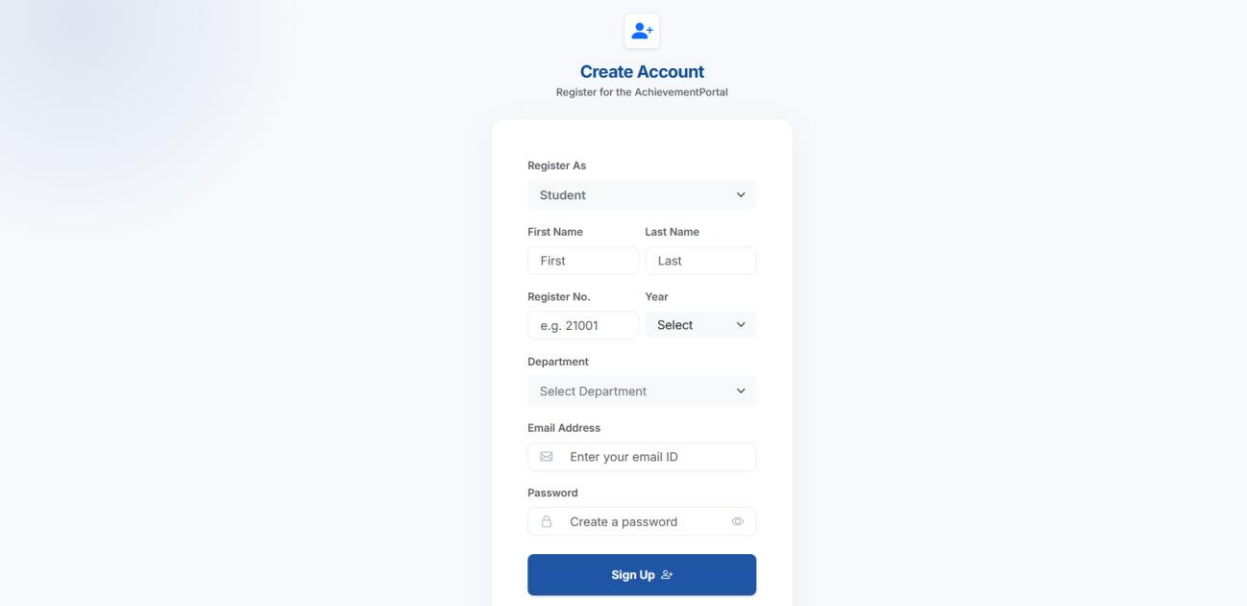
Following successful credential verification, the system performs a secondary authorization check. The role selected in the dropdown determines which database table, namely students, faculty, or admins, is queried using the maybeSingle() method to confirm that the authenticated user's identifier exists within that specific table. This two-step process ensures that possession of valid login credentials alone is insufficient to gain access; the user must also be a legitimate member of the role they have claimed. If the user's identifier is not found in the corresponding role table, the system explicitly invokes supabase.auth.signOut() to terminate the newly created session and raises an "Access Denied" error, thereby preventing a user from masquerading under a role to which they do not belong.

Once both credential verification and role verification succeed, the system persists the access_token and the userRole value in the browser's localStorage. This locally stored state is subsequently used by the route protection mechanism to determine whether a user is permitted to view a particular page without requiring a fresh authentication check on every navigation event.

1.4 Technical Implementation

The authentication functionality is implemented primarily within src/Login.jsx and src/Signup.jsx, supported by src/components/ProtectedRoute.jsx for route-level enforcement and src/supabaseClient.js for establishing the connection to the Supabase backend. The ProtectedRoute component intercepts navigation attempts to private routes by first calling supabase.auth.getSession() to confirm that an active Supabase session exists. If a session is present, the component cross-references the requiredRole prop, which is supplied to it by the parent route configuration, against the userRole value stored in localStorage. Should these values fail to match, or should no valid session be found at all, the component redirects the user back to the /login page.

Logout functionality is distributed across the layout components used by each role, namely Layout.jsx for students, FacultyLayout.jsx for faculty, and AdminLayout.jsx for administrators. Each of these invokes a handleLogout function that calls supabase.auth.signOut(), clears the relevant entries from localStorage, and navigates the user back to the root route. The data flow across the authentication process can be summarized as follows: credentials are submitted to the Supabase Auth API; a JSON Web Token is returned and the user.id is extracted from the resulting session; this identifier is used to query the students, faculty, or admin table as appropriate; upon successful verification, the JWT and role are stored locally; and React Router DOM subsequently redirects the user to the corresponding dashboard. The implementation relies on React Hook state through useState and useEffect, the Supabase Auth client from the @supabase/supabase-js package, React Router DOM utilities including useNavigate and Navigate, and the browser's native localStorage API.



1.5 Advantages

The dual-verification approach, combining Supabase's credential authentication with an explicit role-table lookup, provides a meaningful layer of authorization beyond simple login validation, reducing the likelihood that a user could access functionality reserved for another role. Centralizing authentication through a single login entry point simplifies the user experience while keeping the underlying role logic consistent across the application. The use of localStorage for session and role persistence allows the application to avoid redundant authentication checks on every page transition, improving perceived responsiveness. Furthermore, explicit sign-out on role mismatch ensures that no partially authenticated state is left active in the browser, closing a potential security gap.

1.6 Conclusion

The authentication module of the College Portal establishes a robust and role-aware access control mechanism by combining Supabase Auth's credential verification with a secondary, table-based role confirmation step. This design ensures that the correct dashboard and privileges are made available to each category of user while actively preventing unauthorized escalation of access. Supported by consistent route protection through the ProtectedRoute component and

clean session termination during logout, the authentication subsystem forms a dependable foundation upon which the remainder of the portal's role-specific functionality is built.

2. Dashboard Working

2.1 Introduction

The dashboard represents the primary operational interface through which each category of user interacts with the College Portal after successful authentication. Given that the portal supports three distinct roles, the application implements three functionally separate dashboards, one each for students, faculty, and administrators, with each dashboard tailored to surface the specific data and controls relevant to its respective user type. This separation ensures that the information density and available actions on each dashboard correspond directly to the responsibilities of the role viewing it.

2.2 Purpose

The purpose of the dashboard components is to provide each user with a consolidated, at-a-glance view of the metrics and actions most relevant to their role, thereby reducing the need to navigate through multiple pages to obtain routine information. For students, this means visibility into personal performance and achievement status; for faculty, it means efficient oversight of pending student submissions; and for administrators, it means centralized control over institutional configuration such as departments, achievement categories, and reward rules. In each case, the dashboard functions as the operational hub from which further role-specific actions are initiated.

2.3 Working Principle

Each dashboard follows a common underlying principle: upon mounting, the component calls supabase.auth.getUser() to retrieve the identity of the currently authenticated user, and then uses the returned user.id to query the relational tables associated with that role. The retrieved data is stored in React state arrays, such as achievements, pendingList, or departments, which are subsequently mapped to corresponding JSX elements for rendering. This pattern of fetch-then-render is applied consistently across all three dashboards, though the specific tables and computations differ according to role.

On the Student Dashboard, implemented in Dashboard.jsx and StudentDashboard.jsx, the system displays the total reward points accumulated by the student, their rank on a global leaderboard, counts of approved and pending achievements, and a dynamically computed tier badge. The badge, which may be Bronze, Silver, Gold, or Platinum, is calculated from the student's cumulative total_points value using the getBadgeForPoints utility function. The dashboard

additionally fetches department information by joining the students and departments tables, and retrieves a preview of recent achievements through the fetchAchievements utility.

The Faculty Dashboard, implemented within FacultyLayout.jsx, presents counts of pending approvals and verified achievements alongside two principal tables: a Pending Approvals table and a Verified Students table. Faculty members are able to approve or reject student submissions directly from this interface. When an achievement is approved, the system updates the achievement's status, inserts a corresponding record into the reward_points table, increments the student's total_points value in the students table, and inserts a push notification into the notifications table, thereby propagating the effect of a single approval action across multiple related tables.

The Admin Dashboard, implemented within AdminLayout.jsx, surfaces aggregate statistics including the total number of departments, students, faculty, and active reward rules. It further embeds tabbed interfaces for managing Departments through create, read, update, and delete operations on the departments table, managing Achievement Categories through similar operations on the achievement_categories table, and configuring Reward Rules by updating the base_points field within achievement_categories. The Admin Dashboard also includes functionality for sending global announcement notifications.

2.4 Technical Implementation

The dashboard functionality spans several files, including src/pages/Dashboard.jsx, src/pages/StudentDashboard.jsx, src/layouts/FacultyLayout.jsx, src/layouts/AdminLayout.jsx, and src/utils/achievementUtils.js, with administrative sub-functionality further distributed across dedicated components such as Departments.tsx, AchievementCategories.tsx, and RewardRules.tsx. Database tables involved across the three dashboards include students, faculty, departments, achievements, achievement_categories, and reward_points. Key functions include getBadgeForPoints for tier calculation, handleVerificationAction for the faculty approval workflow, and fetchData, which serves as the general-purpose data loader invoked by the dashboards on mount. The dashboards are built using React Hook state through useState and useEffect, the Supabase PostgREST client for data retrieval, and a combination of Lucide React and Bootstrap Icons for visual presentation.

2.5 Advantages

Separating dashboard logic by role allows each interface to remain focused and uncluttered, presenting only the information and actions relevant to the user viewing it. The consistent fetch-

then-render pattern across all three dashboards contributes to a predictable and maintainable codebase, as new metrics or tables can be added following an established convention. The cascading update logic triggered by faculty approval, whereby a single action updates achievement status, reward points, student totals, and notifications, ensures that related data remains synchronized across the system without requiring the user to perform multiple manual steps. The administrative tab-based configuration interface further consolidates institutional management tasks into a single, cohesive dashboard.

2.6 Conclusion

The dashboard subsystem of the College Portal successfully delivers role-appropriate visibility and control to students, faculty, and administrators through three dedicated interfaces built on a shared architectural pattern of authenticated data retrieval and state-driven rendering. By tailoring statistics, tables, and available actions to each role, and by ensuring that interdependent data such as points, statuses, and notifications remain synchronized following key actions like faculty approval, the dashboards collectively serve as the functional core of the application's day-to-day usage.

3. Responsive Layout

3.1 Introduction

Responsive design refers to the practice of constructing user interfaces that adapt their layout and presentation according to the size and characteristics of the device on which they are viewed. Given that users of the College Portal may access the system from desktop computers, tablets, or mobile phones, it is necessary for the application's interface to remain usable and visually coherent across this range of screen sizes. The College Portal achieves this through a combination of an established CSS framework and custom styling.

3.2 Purpose

The purpose of implementing responsive design in the College Portal is to ensure that all users, irrespective of the device they employ, can comfortably view dashboards, navigate menus, and interact with forms without encountering broken layouts, overlapping elements, or inaccessible controls. This is particularly important in an academic context where students, faculty, and administrators may need to check statuses or perform approvals while away from a desktop workstation.

3.3 Working Principle

The responsive behavior of the application is governed principally by Bootstrap 5, which is used extensively for its grid system, spacing utilities such as mb-3 and p-4, and flexbox utility classes such as d-flex and justify-content-between. Layout components make use of Bootstrap's breakpoint system, applying column classes such as col-12 col-md-6 col-xl-4 so that content transitions from a single-column arrangement on mobile devices to a multi-column arrangement on larger desktop viewports. Flexbox utility classes, including flex-column, flex-md-row, align-items-center, and gap-3, are further employed to keep headers, cards, and forms fluidly aligned across varying screen widths.

Beyond the Bootstrap framework, custom CSS defined in files such as App.css and Login.css, along with custom classes such as glass-panel and fade-in-up, supplement the responsive grid with a distinctive visual aesthetic. On smaller screens, the sidebar within Layout.jsx operates as an offcanvas menu, with its visibility controlled by the sidebarOpen state and toggled through CSS classes involving translate-x or absolute positioning. Navigation bars across the application incorporate hamburger menu icons, specifically bi-list, which toggle the sidebar or collapse navigation links once the viewport width falls below the md or lg breakpoint thresholds.

3.4 Technical Implementation

The responsive layout implementation is distributed across src/App.css, src/index.css, src/Login.css, src/components/Layout.css, and src/components/Layout.jsx. Reusable UI components contribute to the consistency of the responsive design: glassmorphism-styled panels implemented through the glass-panel, glass-input, and glass-table classes provide a unified premium aesthetic across multiple dashboards, while Bootstrap's badge, rounded-pill, and contextual color classes such as bg-primary-subtle and text-primary are reused throughout the application for tags and status indicators. The overall styling approach relies on Bootstrap 5 for its CSS grid and utility classes, custom vanilla CSS for bespoke visual treatments, and CSS custom properties such as --primary-brand for maintaining consistent theming across components.

3.5 Advantages

Leveraging Bootstrap 5's grid and utility classes allows the application to achieve broad device compatibility without the overhead of writing extensive custom media queries for every component. The combination of Bootstrap breakpoints with flexbox utilities provides fine-grained control over element alignment across screen sizes, while the offcanvas sidebar pattern preserves navigational access on mobile devices without permanently consuming screen space. The reuse of custom classes such as glass-panel and badge-related utilities across multiple dashboards ensures visual consistency throughout the application, reducing the cognitive load on users as they move between different sections of the portal.

3.6 Conclusion

The responsive layout strategy adopted in the College Portal, combining the structural capabilities of Bootstrap 5 with custom CSS refinements, enables the application to present a coherent and usable interface across desktop and mobile devices alike. Through the deliberate use of breakpoints, flexbox alignment, and an adaptive offcanvas sidebar, the portal ensures that its functionality remains accessible regardless of the device on which it is accessed, while reusable styling components maintain a consistent visual identity throughout the system.

4. Navigation

4.1 Introduction

Navigation within a web application refers to the mechanisms by which users move between different views or pages without requiring the browser to reload the entire application. The College Portal implements client-side routing to manage this movement efficiently, while simultaneously enforcing a strict separation between routes accessible to unauthenticated visitors and routes that require a verified role-based session.

4.2 Purpose

The purpose of the navigation architecture is twofold: first, to provide a smooth and responsive browsing experience by avoiding full page reloads as users move between sections of the portal, and second, to enforce access control at the routing level so that users cannot reach pages or dashboards belonging to roles other than their own simply by manipulating the URL. This ensures that navigation itself becomes an extension of the authentication and authorization mechanisms already established within the application.

4.3 Working Principle

The routing structure of the College Portal is configured using the BrowserRouter component from React Router DOM within App.jsx, where explicit Route components map individual URL paths to their corresponding React components. A fallback wildcard route, defined with path="*", ensures that any unrecognized URL is redirected back to the /login page rather than resulting in a broken or blank view. Protected sections of the application are wrapped by the ProtectedRoute component, which accepts a requiredRole prop and verifies it against both the localStorage-stored role and the active Supabase session before allowing the nested Outlet to render.

Navigation within each role's interface follows a slightly different pattern depending on the complexity of that role's functionality. For students, navigation is managed by the Layout component, which maintains a persistent sidebar containing NavLink elements from react-router-dom that map to distinct URL paths such as /dashboard, /my-achievements, and /reward-points. For faculty and administrators, navigation is instead managed internally within FacultyLayout.jsx and AdminLayout.jsx respectively; rather than routing to separate URL paths for each sub-section, these layouts maintain an internal state variable, such as currentTab, which

is set according to location.pathname and used to dynamically render the appropriate sub-component while preserving the surrounding layout shell.

4.4 Technical Implementation

The navigation logic is implemented primarily within src/App.jsx, src/components/ProtectedRoute.jsx, and src/components/Layout.jsx, drawing on React Router DOM v6 components including Router, Routes, Route, Navigate, Outlet, NavLink, and Link. The overall page hierarchy places the Login component at the root path /, with the student experience wrapped in the Student Layout and accessible at /dashboard, the faculty experience handled by the FacultyLayout tab engine at /faculty-dashboard, and the administrative experience handled by the AdminLayout tab engine at /admin-dashboard. This hierarchy reflects the underlying role separation established during authentication and carried through into every subsequent navigational decision made by the application.

4.5 Advantages

Client-side routing through React Router DOM allows the College Portal to transition between views rapidly, without incurring the latency and visual disruption associated with full page reloads. Wrapping protected routes with the ProtectedRoute component centralizes access control logic in a single, reusable location rather than duplicating role checks across individual pages. The hybrid navigation approach, using distinct URL paths for student pages while relying on internal tab state for faculty and administrative sub-sections, allows each role's interface to be structured in the manner best suited to its complexity, with the tab-based approach in particular reducing the number of discrete routes that must be defined and protected for the more feature-dense faculty and admin dashboards.

4.6 Conclusion

The navigation architecture of the College Portal, built upon React Router DOM v6 and reinforced by the ProtectedRoute component, provides both a fluid single-page application experience and a consistent enforcement of role-based access at the routing level. By combining explicit route definitions for authentication and student pages with internal tab-driven navigation for the more complex faculty and administrative layouts, the system achieves an appropriate balance between routing clarity and interface flexibility.

5. Forms

5.1 Introduction

Forms serve as the primary mechanism through which users of the College Portal submit data to the underlying Supabase database, whether that data pertains to authentication credentials, newly registered accounts, achievement submissions, or administrative configuration changes. The application implements its forms as controlled React components, meaning that the value of every input field is bound directly to React state and updated through explicit event handlers.

5.2 Purpose

The purpose of the form implementations across the College Portal is to provide structured, validated entry points for the various categories of data the system must collect, ranging from login credentials and new account registrations to student achievement submissions and administrative department records. Each form is designed around the specific data requirements and downstream database operations associated with its function, ensuring that user input is captured accurately before being persisted through the Supabase client.

5.3 Working Principle

All forms in the application follow the controlled component pattern, in which each input field's current value is derived from React state and any change to that field triggers a state update through an onChange handler. Submission is handled asynchronously through onSubmit handlers that invoke the Supabase client, with the resulting promise awaited so that the outcome of the operation can be reflected in the user interface, typically through loading indicators, error banners, or success messages.

The Login form, implemented in Login.jsx, collects a role selection, an email address, and a password, maintaining these alongside loading and error state. Validation is performed through HTML5 required attributes, and submission invokes supabase.auth.signInWithPassword before verifying the existence of the corresponding role-table record and storing the resulting JSON Web Token. Errors are surfaced through alert banners driven by authError or roleError state values.

The Signup form, implemented in Signup.jsx, captures first name, last name, email, role, and password within a formData object, alongside loading, error, and success state. Upon submission, the form calls supabase.auth.signUp(), and on success executes sequential INSERT queries into the public.users table and into the specific role table corresponding to the selected role, namely

students, faculty, or admin. A success message is rendered prior to redirecting the user to the /login page.

The Student Achievement form, implemented in StudentAchievementForm.jsx, is the most data-intensive form in the application, collecting a title, a category selected from a dropdown, a description, certificate and document file uploads, and fallback URL inputs, with corresponding state for formData, certificateFile, documentFile, categories, submitting, error, and success. Upon submission, the form uploads the selected files to the certificates bucket in Supabase Storage using supabase.storage.from('certificates').upload(), retrieves the resulting public URL through getPublicUrl(), and inserts the complete submission payload into the achievements table with an initial status of Pending. The form then queries the student's department_id, identifies matching faculty members from the faculty table, and performs a bulk insert of notification records into the notifications table for those faculty members. The form disables its inputs during submission, renders a success interface upon completion, and redirects the user to the dashboard after a short delay.

The Admin Departments form, implemented within Departments.tsx, is presented in a modal and collects a department name and description, maintained through newDeptName, newDeptDesc, and showAddModal state. Upon submission, the form inserts the new department record into the departments table via Supabase, retrieves the returned object, and appends it to the local React state array so that the updated list of departments is reflected immediately in the interface without requiring a full data refresh.

5.4 Technical Implementation

The form-related functionality is distributed across src/Login.jsx, src/Signup.jsx, src/pages/StudentAchievementForm.jsx, src/admin/components/Departments.tsx, src/admin/components/AchievementCategories.tsx, and src/admin/components/RewardRules.tsx. All forms rely on React's controlled input pattern together with onSubmit event handlers, and the more data-intensive forms, such as the Student Achievement form, additionally make use of the Supabase Storage API for handling file uploads alongside standard PostgREST insert operations for persisting structured data.

5.5 Advantages

The consistent use of controlled components across all forms in the application ensures predictable state management and simplifies validation and error handling. Structuring form-related state around dedicated objects, such as formData in the Signup and Student Achievement

forms, keeps related fields organized and reduces the proliferation of individual state variables. The Student Achievement form's automatic notification of relevant faculty members upon submission removes the need for students to manually alert faculty of a new submission, while the Admin Departments form's immediate local state update after insertion provides responsive feedback without requiring a full page or data reload.

5.6 Conclusion

The forms implemented throughout the College Portal collectively provide a consistent, controlled, and validated means of capturing user input across authentication, registration, achievement submission, and administrative configuration workflows. By pairing React's controlled component pattern with asynchronous Supabase operations, including database insertions and file storage uploads, each form reliably translates user-provided data into persisted records while providing appropriate feedback throughout the submission process.

6. Basic API Integration

6.1 Introduction

The College Portal's connection to its backend data store is realized entirely through the official @supabase/supabase-js client library, which communicates with the underlying PostgreSQL database via the PostgREST API layer exposed by the Supabase platform. This client-side integration underlies every data-driven feature of the application, from authentication to dashboard rendering to form submission.

6.2 Purpose

The purpose of the API integration layer is to provide a consistent, promise-based interface through which every component of the application can authenticate users, retrieve records, and persist changes to the database, without requiring components to manage low-level HTTP requests directly. Centralizing this configuration ensures that all parts of the application communicate with the same Supabase project instance using a uniform set of credentials and conventions.

6.3 Working Principle

The Supabase client is configured once, within supabaseClient.js, using the createClient() function, which consumes the environment variables VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY to establish a connection to the correct Supabase project. Once instantiated, this client exposes both the authentication API, used for operations such as signInWithPassword(), signUp(), getUser(), getSession(), and signOut(), and the PostgREST-based data API, used for SELECT, INSERT, UPDATE, and DELETE operations against the underlying database tables.

SELECT queries are constructed through method chaining, following the general pattern of supabase.from('table').select('columns').eq('field', value). The application makes use of Supabase's foreign key joining capability in queries such as retrieving student details through .from('students').select('first_name, last_name, year, departments(name)').eq('user_id', targetId).single(), which resolves related department information within a single query. List-oriented queries, such as retrieving notifications through .from('notifications').select('*').eq('user_id', user.id).order('created_at', { ascending: false }), similarly rely on this chained method syntax.

INSERT operations are used both within forms and as part of automated workflows triggered by user actions. Examples include submitting an achievement through .from('achievements').insert(achievementPayload), distributing reward points through .from('reward_points').insert([{ student_id: id, points_awarded: val }]), and broadcasting administrative announcements by passing an array of notification objects directly to .insert(inserts) for bulk insertion. UPDATE operations handle inline edits and status transitions, such as the faculty approval workflow implemented through .from('achievements').update({ status: 'Approved', approved_by: user.id }).eq('id', id), the corresponding update of a student's accumulated points through .from('students').update({ total_points: newTotal }).eq('user_id', studentId), and marking notifications as read through .from('notifications').update({ is_read: true }).eq('user_id', user.id). DELETE operations are used within the My Achievements page to allow removal of pending submissions, following the pattern .from('achievements').delete().eq('id', id).

6.4 Technical Implementation

The API integration is anchored by src/supabaseClient.js, with query and mutation logic distributed across src/utils/achievementUtils.js and the various page and component files that house the corresponding useEffect hooks and event handlers. The tables accessed through this integration layer include students, faculty, users, departments, achievements, achievement_categories, reward_points, and notifications. Nearly all queries are wrapped in try...catch blocks, with the error object explicitly destructured from the Supabase response, const { data, error } = ..., and thrown when present so that it can be caught and used to update a local error state string for display to the user. Boolean loading and submitting states are set to true at the beginning of each asynchronous function and reset to false within a finally block, ensuring that loading indicators and disabled UI states are cleared regardless of whether the operation succeeded or failed. The integration is built on @supabase/supabase-js version 2 and relies throughout on JavaScript Promises and async/await syntax.

6.5 Advantages

Centralizing the Supabase client configuration within a single file ensures that every part of the application communicates with the backend using consistent credentials and settings, simplifying maintenance and reducing the risk of misconfiguration. The chained query syntax offered by the PostgREST client allows complex filtering, ordering, and relational joins to be expressed concisely without hand-written SQL. Consistent use of try...catch blocks combined with finally-based loading state resets ensures that the interface reliably communicates operation outcomes to the

user and never leaves loading indicators active indefinitely, even in the event of a network or database error.

6.6 Conclusion

The Basic API Integration layer of the College Portal, built entirely upon the @supabase/supabase-js client and the PostgREST endpoints it exposes, provides a consistent and centralized means of performing authentication, data retrieval, and data mutation across the entire application. Through disciplined use of chained queries, structured error handling, and predictable loading state management, this integration layer supports every other subsystem described in this report, from authentication and dashboards to forms and navigation, and forms the backbone through which the College Portal interacts with its underlying data.